EE5203 L01 Practice Worksheet

Toolchain reasoning, prediction, and debugging

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Expected time: 30-45 minutes
Policy: Attempt every question before consulting the worked solutions.

Part A - Core concepts

1. For each item, classify it as a **processor**, **microcontroller**, **peripheral**, **development board**, or **embedded system**. Some descriptions may contain more than one relevant level; state the most useful classification.
   a. Arm Cortex-M4
   b. STM32F401RE
   c. Nucleo-F401RE
   d. TIM2
   e. A thermostat containing an STM32 board, temperature sensor, and relay
2. Give one example of each constraint:
   a. timing constraint;
   b. memory constraint;
   c. energy constraint;
   d. safety or reliability constraint.
3. A project builds with zero errors, but the LED never changes. Explain why this is not a contradiction.

Part B - Toolchain sequencing

4. Put these actions in a valid order:
   - flash firmware;
   - link object files;
   - edit `main.c`;
   - compile source files;
   - observe the LED;
   - inspect a variable at a breakpoint.
5. Match each symptom to the most likely initial investigation.

Symptom	Investigation choices
`expected ';' before '}'`	A. Debug connection/USB
`undefined reference to process_sample`	B. Compile-time syntax
`No ST-LINK detected`	C. Link-time definition
Program pauses but variable is wrong	D. Run-time logic/state

6. What evidence would you record for each claim?

    a. The source compiled.

    b. The firmware was written to the board.

    c. The loop executed repeatedly.

    d. The physical output met the timing requirement.

Part C - Predict and trace

7. Trace the code without running it.

```
int count = 0;

while (count < 3) {
    count = count + 1;
}
```

a. List `count` after each loop body execution.

b. What is its final value?

c. On which line would you place a breakpoint to observe the value changing?

8. Predict the observable difference between the two loops.

```
/* Program A */                /* Program B */
while (1)                      while (1)
{                              {
    HAL_GPIO_TogglePin(...);       HAL_GPIO_TogglePin(...);
    HAL_Delay(500);                HAL_Delay(100);
}                              }
```

What remains identical? What changes? What measurement would support your claim?

9. A student changes `HAL_Delay(500)` to `HAL_Delay(100)`, presses **Run**, and observes no change. List four distinct explanations and give one test for each.

Part D - Debugging decisions

10. Rewrite "My code does not work" as a useful debugging report using:

    - expected behavior;
    - observed behavior;
    - evidence;
    - next test.

11. A breakpoint placed on `HAL_GPIO_TogglePin(...)` is never reached. Rank the following checks in a sensible order and justify the first two:

    - confirm the correct project was flashed;
    - replace the USB cable;
    - confirm the program reached `main`;
    - inspect whether execution is paused elsewhere;
    - check the loop and branch conditions.

12. The LED blinks correctly, but the debugger displays an unexpected local variable value. Give two reasons the LED result alone cannot prove the variable is correct.

Part E - Project connection

13. Choose one potential project and complete this chain:

```
input -> acquisition -> processing -> output -> communication
```

14. Define the **minimum viable version** in no more than three sentences. Then name one optional extension that should only begin after the minimum system works.

Exit self-assessment

Mark one:

- ☐ Ready for the L01 lab without additional review
- ☐ Need to review the toolchain path
- ☐ Need help with C control flow
- ☐ Need help with the VS Code / CubeMX setup